

Hormone messages: the control room

Teacher guide and worked answers

Describe endocrine signalling and explain why suitable internal conditions matter.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Summer | C30

Describe hormonal coordination & homeostasis.

Directly develops C30 through Pearson 7.1, 7.9 and selected 7.10B concepts. Core names endocrine glands and blood transport, with pancreas/insulin as a simple example. Detailed adrenaline, thyroxine feedback and reproductive hormone mechanisms are not claimed as covered.

Prior learning

Blood transports substances around the body.

Cells carry out reactions using enzymes.

External conditions can change while an organism remains alive.

Success evidence

Trace gland → blood → target organ.

Explain why a circulating hormone does not affect every cell equally.

Explain homeostasis as regulation within suitable limits rather than perfect stillness.

Equipment

Shared evidence sheet, one selected response route, pen and ruler.

Projector or offline HTML/PowerPoint; calculator where useful.

Preparation

Print the shared page, one selected response route and the exit page. Routes are alternatives, not extra tasks.

Read the worked answers and note any constructed data before teaching.

Practical care

Paper and screen lesson; no human sampling, exercise test, breath holding, medicine handling or personal health disclosure.

Ready to teach alternative

All evidence is supplied. Use paper diagrams and cards, pointing, annotation or dictation if a device is unavailable.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	How does a cell receive the right message?
2	Retrieval	2-5	Transport and conditions
3	I do	5-9	I trace the message route
4	I do	9-12	Recognition makes the signal specific
5	I do	12-15	Homeostasis keeps conditions suitable
6	We do	15-18	Message-routing desk
7	We do	18-20	Predict the responding cell
8	We do	20-22	Control-room repair
9	Hinge	22-25	Which explanation links route and recognition?
10	You do	25-26	Build a control explanation
11	You do	26-36	Make the mechanism visible
12	Review	36-38	Audience asks about the missing link
13	Review	38-39	A useful model also has a limit
14	Exit	39-40	Three linked ideas
15	Exit	Within stage	Choose the next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening How does a cell receive the right message?

Invite a prediction without calling hormones thinking agents. Preview signal route, recognition and control.

2 Retrieval Transport and conditions

R1: blood. R2: cell reactions/enzymes work effectively under suitable conditions. R3: no; interactions are physical/chemical. Accept plain-language mechanisms.

3 I do I trace the message route

Model pancreas releasing insulin into blood; responsive tissues change glucose uptake/storage processes. Keep this preview qualitative. Endocrine secretion is into blood, not an exocrine duct.

4 I do Recognition makes the signal specific

Think aloud: "Both cells are exposed, so travel alone cannot explain the difference. Cell A has the appropriate receptor. The simplified shape represents molecular recognition, not a key operated by a tiny person."

5 I do Homeostasis keeps conditions suitable

Model a raised internal variable and a response tending to lower it. Do not use universal numeric normal ranges. Link suitable temperature to enzyme activity and water balance to cell volume at a conceptual level.

6 We do Message-routing desk

Use table. All six locations are curriculum gland knowledge; reproductive organs are named matter-of-factly without requiring personal information. Route pancreas → blood → appropriate target tissue.

7 We do Predict the responding cell

A has matching receptor. Ask for the missing link if pupil says hormone never reaches B; it does reach B but no matching response is represented.

8 We do Control-room repair

It counteracts the original rise, tending towards suitable limits. Do not claim every biological loop is negative feedback or control is instantaneous.

Reveal: A negative-feedback response opposes the departure, rather than automatically amplifying it.

9 Hinge Which explanation links route and recognition?

First option correct. Re-teach blood distribution versus recognition if wrong.

A. The hormone circulates; appropriate target cells recognise it and respond. - Yes. Transport and recognition are different parts of the mechanism.

B. The hormone chooses only the cells it likes. - Hormones do not choose; molecular recognition and cell machinery determine responses.

C. Homeostasis means no internal value ever changes. - Regulation responds to changes and helps keep conditions within suitable limits.

10 You do Build a control explanation

All routes target message transport, specificity and regulated conditions. No numerical clinical data are required.

11 You do Make the mechanism visible

Supported S1-S4; standard M1-M4; stretch T1-T4. Use a word-bank prompt if needed and record support. Do not require advanced hormone feedback cascades.

12 Review Audience asks about the missing link

A quiet teacher/TA review counts. Pupil improves a route label or distinguishes exposure from response.

13 Review A useful model also has a limit

Model a precise limit. Keep homeostasis as dynamic regulation; avoid perfect thermostat equivalence.

14 Exit Three linked ideas

E1: in blood. E2: appropriate recognition/receptor absent in the model. E3: internal conditions within suitable limits. Use these to prepare the glucose model.

15 Exit Choose the next useful step

Shared exit minute. Accept a labelled arrow chain instead of extended writing.

Answers and assessment

Retrieval and We do

R1: blood; R2: suitable conditions support cell reactions/enzymes and cell integrity; R3: no. W1: use the six locations in the table, then pancreas → blood → responsive tissue. W2: A has matching recognition. W3: the response should reduce the original rise in the model.

Hinge

Correct: circulation plus target recognition. Hormones do not consciously choose. Homeostasis permits change while regulating within suitable limits.

S1 / M1

An endocrine gland releases a hormone into blood. Circulating blood transports it to tissues. Cells with suitable recognition/response machinery respond by changing activity.

S2 / M2

A has the matching receptor; B does not in this model. Both can encounter the hormone. Presence in blood is not the same as an appropriate target response.

S3 / M3

A corrective response opposes the rise and tends towards suitable limits. Homeostasis regulates an internal environment; it does not mean every value is fixed or that all coordination is hormonal.

S4 / M4

Pancreas upper abdomen; thyroid neck. Other valid locations: pituitary base of brain; adrenals above kidneys; ovaries pelvis; testes scrotum. Suitable temperature supports enzyme-controlled reactions; water balance prevents excessive cell swelling/shrinking.

T1

No. Recognition is one requirement; downstream cell machinery must also function. The simple receptor picture omits those steps.

T2

No. Delays and overshoot can occur in regulated systems. A simplified diagram does not show response speeds, competing signals or exact ranges.

T3

Hormones circulate chemically in blood and affect responsive targets. Nerve impulses travel along neurones to connected targets. Typical timing/range differ, but avoid absolute claims that every hormone is slow or every response identical.

T4

The model has no individual measurements, calibrated parameters or clinical context. It illustrates a causal organisation, not a diagnostic instrument.

Exit and assessment

E1: bloodstream. E2: lack of appropriate receptor/response machinery; use the receptor distinction for core. E3: internal conditions within suitable limits so cells function. Secure core: correct route+specificity+dynamic regulation. Re-teach by separately asking “does the signal arrive?” and “can the cell respond?”

Teaching and assessment notes

40-minute lesson. Zero-minute slides share the preceding stage allocation.

Choose one response route. Accept accurate speech, pointing with explanation, annotations or writing; record the support used.

A teacher or TA can be a quiet audience: ask one evidence question, then let the pupil revise or justify a decision.

Suggested timing: 2+3+10+7+3+11+3+1=40 minutes.

Interactive We do uses the same gland-route, target-recognition and feedback cases. Pupils predict, reveal a mechanism and explain; paper W1-W3 follows the same decisions.

Naming ovaries and testes is part of locating endocrine structures; no pupil anatomy, identity, family history or health disclosure is required.

Access and responsive teaching

Supported

Same core concepts with chunked prompts and read-aloud access. Record support used; a pathway is not a diagnosis or fixed attainment label.

Standard

Explain the supplied evidence independently using accurate scientific terms and units.

Stretch

Optional deeper transfer or evaluation. Higher-tier-only material is labelled explicitly and is not required for the core outcome.

Regulation

Offer solo or partner work, a quiet response, a pass and brief reset with clear re-entry. No forced disclosure or public performance.

Misconceptions to check

A hormone travels only through a private tube to one organ.

It circulates in blood; target cells respond because they can recognise the signal.

Check: Ask why a nearby non-target cell may not respond.

Every cell responds identically to every hormone.

Cells differ in receptors and response machinery.

Check: Compare the two target-cell models.

Homeostasis means a condition never changes.

Regulation limits departures; suitable ranges and dynamic adjustment matter.

Check: Ask whether a small change automatically means control has failed.

All homeostatic responses must use hormones.

Homeostasis can involve nervous and hormonal coordination.

Check: Distinguish the general control idea from the endocrine example.

Word	Meaning
hormone	A chemical messenger released by an endocrine gland and carried in blood.
endocrine gland	An organ or tissue that releases hormones into the bloodstream.
target cell	A cell able to respond to a particular hormone, usually because it has the appropriate receptor.
receptor	A molecule that recognises a signal and can trigger a response.
homeostasis	Regulation of internal conditions within suitable limits despite change.
negative feedback	A response that reduces the original departure from a regulated condition.

Scientific references

[Pearson Edexcel GCSE Biology 1B10 specification](#)

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 9 September 2026.

[NIDDK: What is diabetes?](#)

Checked the pancreas-insulin example. No diagnostic threshold, medicine choice or treatment advice is taught. Checked 9 September 2026.

[OpenStax Anatomy and Physiology2e: An overview of the endocrine system](#)

Checked endocrine transport and target-cell specificity; diagrams and classroom cases are original. Checked 9 September 2026.